



HOMEWORK ASSIGNMENT

MICROECONOMIC ANALYSIS

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QUESTION K2.1

Let $>$ be a binary relation on a set X , and define $\geq$ as in section 2.1; $x \geq y$ if it is not the case that $y > x$. Show that $\geq$ is asymmetric if and only if $>$ is complete. Show that $>$ is negatively transitive if and only if $\geq$ is transitive.

Let $>$ be a binary relation on a set X , and define the associated weak relation $\geq$ by the rule:

$$x \geq y \text{ iff } \neg(y > x)$$

Completeness of $>$ implies asymmetry of $\geq$

The first claim is that $>$ is complete if and only if $\geq$ is asymmetric. We must recall that a relation R is asymmetric when xRy always implies $\neg(yRx)$, and that $>$ is complete when for every pair x, y either $x > y$ or $y > x$.

We are going to assume first that $\geq$ is asymmetric. Choose any x, y . If $y > x$ then completeness is satisfied. If instead y not $> x$, then by definition $x \geq y$. Asymmetry then gives $\neg(y \geq x)$.

But the statement $y \geq x$ is equivalent to $\neg(x > y)$, so $\neg(y \geq x)$ is equivalent to $x > y$. Therefore in this case we can also obtain one of the strict comparisons, and so for all pairs we have either $x > y$ or $y > x$, in consequence $>$ is complete.

Now assuming that $>$ is complete. Take any x, y such that $x \geq y$, which by definition means that y not $> x$. Completeness leaves only one remaining possibility, namely $x > y$. The statement $y \geq x$ is simply the negation of $x > y$, so from $x > y$ we obtain $\neg(y \geq x)$. As a consequence $x \geq y$ implies $\neg(y \geq x)$, which is exactly the condition of asymmetry for $\geq$.

Negative transitivity of $\geq$ implies transitivity of $>$

The second claim is that $\geq$ is negatively transitive if and only if $>$ is transitive. Negative transitivity means that for all x, y, z :

$$\neg(x \geq y) \text{ and } \neg(y \geq z) \Rightarrow \neg(x \geq z).$$

A key observation is that since $u \geq v$ means $\neg(v > u)$, its negation becomes:

$$\neg(u \geq v) \Leftrightarrow v > u$$

Therefore each negation of a weak preference simply flips into a strict preference in the opposite direction. We are going to suppose first that $\geq$ is negatively transitive, and take $a > b$ and $b > c$. Rewriting these as $\neg(b \geq a)$ and $\neg(c \geq b)$, negative transitivity yields $\neg(c \geq a)$, which again is the same as $a > c$. Hence transitivity of $>$ follows.

For the reverse direction, assume $>$ is transitive. If $\neg (x \geq y)$ and $\neg (y \geq z)$, then as noted above these statements are equivalent to $y > x$ and $z > y$. By transitivity of $>$ we obtain $z > x$. Turning this back into a statement about $\geq$, we find $z > x \Leftrightarrow \neg (x \geq z)$. In consequence the defining condition of negative transitivity of $\geq$ holds.

The hint in the problem suggests noticing that negative transitivity is simply the contrapositive form of ordinary transitivity of $>$. Indeed,

$$x > y \text{ and } y > z \Rightarrow x > z$$

Is logically equivalent to

$$\neg (x > z) \Rightarrow \neg (x > y) \text{ or } \neg (y > z)$$

And when rewritten in terms of $\geq$ this becomes exactly the formulation of negative transitivity given in the statement of the exercise. This explains why the name “negative transitivity” is used: it is transitivity expressed entirely through negated comparisons.

QUESTION S1.8

Consider a monopolist whose total cost function is $C=kx^2$ and who faces the demand curve $x = a - bp$.

a. What restrictions on the values of the parameters a , b and k would you be inclined to assert a priori?

Before solving the model, we must impose certain restrictions on the demand and cost parameters for maintaining economic reasoning and mathematical consistency:

1. $b > 0$

The parameter b represents the slope of the demand curve, requiring $b > 0$ ensures that the curve is downward sloping, meaning that when price increases, quantity demanded decreases.

2. $a > 0$

The parameter a is the intercept of the demand function, the quantity demanded when the price is zero. Setting it as a positive number ensures that the market size is also positive so the firm can potentially sell output

3. $k \geq 0$

If k is non negative, then costs either stay the same or increase as the firm produces more output. This is a usual assumption: producing more units should never reduce production costs. Assuming the inverse, $k < 0$, would mean that producing more costs less, which is unrealistic.

b. Find the explicit function $x = x^*(t)$. Confirm that for the restricted values of a , b , and k placed in part (a) that $x''(t) < 0$, i.e., output decreases as the tax increases.

We want to find the monopolist's optimal output when a per-unit tax t is imposed. Profit is $\Pi(x) = p(x)x - C(x) - tx$, and since demand is $x = a - bp$, the inverse demand curve is: $p(x) = \frac{a-x}{b}$. The cost function is $C(x) = kx^2$. Substituting this into profit gives:

$$\Pi(x) = \frac{x(a-x)}{b} - kx^2 - tx$$

Now, to maximize the profit, the firm sets marginal profit equal to zero:

$$\frac{d\Pi}{dx} = \frac{a-2x}{b} - 2kx - t = 0$$

This equation tells us that at the optimum, the marginal revenue will be equal to the sum of the marginal cost and tax (marginal revenue = marginal cost + tax).

Calculation process:

$$\frac{a}{b} - \frac{2x}{b} - 2Kx - t = 0 \quad \Rightarrow \quad x \left(\frac{2}{b} - 2K \right) = \frac{a}{b} - t$$

Now we divide both sides:

$$x^*(t) = \frac{\frac{a}{b} - t}{2 \left(\frac{1}{b} + K \right)}$$

Multiplying numerator and denominator by b we obtain the following cleaner expression:

$$x^*(t) = \frac{a + bt}{2(1 + bK)} \quad \rightarrow \quad \text{monopolist's optimal output as a function of the tax}$$

Now we will confirm that for the restricted values of a , b and k , $x^*(t) < 0$. To determine how output responds to a change in the tax, we differentiate this expression with respect to t :

$$x^*(t) = \frac{d}{dt} \left(\frac{a + bt}{2(1 + bK)} \right) = - \frac{b}{2(1 + Kb)}$$

To sign this derivative, we now use the a priori restrictions from answer a imposed on the parameters: $b > 0$ and $k \geq 0$. Given these we observe that:

- The numerator, $-b$, is strictly negative because $b > 0$
- The denominator, $2(1 + bk)$ is strictly positive because $1 + bk \geq 1$ when $k \geq 0$ and $b > 0$

Since a negative number divided by a positive number is negative, we have demonstrated that $x^{*'}(t) < 0$.

c. What restrictions does the hypothesis of profit maximization place on the parameters a , b and k ? Are these weaker or stronger than your a priori restrictions?

Up to this point we have imposed a priori restrictions on the parameters a , b and k to ensure the model was meaningful. Starting from the firm's profit function, the second order condition for a maximum is:

$$\frac{d^2\Pi}{dx^2} = -\frac{2}{b} - 2k < 0$$

And this requires:

$$(i) \frac{1}{b} + k > 0$$

This is the only restriction that profit maximization imposes on the parameters. The condition makes sure that the profit function is concave leaving the interior solution as the maximum. If $b > 0$ and $k \geq 0$ then this last condition automatically holds.

When comparing with our a priori restrictions ($a > 0$, $b > 0$, $k \geq 0$) the profit maximization restriction (i) allows many values of b and k that would be unreasonable, like slightly negative slopes or negative cost coefficients.

The restrictions implied by profit maximization are weaker than the natural economic restrictions. Our a priori assumptions rule out unrealistic parameter values, while the optimization condition alone does not.

d. Substitute your $x^*(t)$ functions into the first-order relation for maximization, and confirms that an identity in t results.

FOC was: $\frac{a-2x}{b} - 2Kx - t = 0$

substituting $x^* = \frac{a-bt}{2(1+bK)}$

Compute $a-2x^*$

$$\begin{aligned} a-2x^* &= a - 2 \left(\frac{a-bt}{2(1+bK)} \right) = a - \frac{a-bt}{1+bK} = \frac{a(1+bK) - (a-bt)}{1+bK} \\ &= \frac{a+abK-a+bt}{1+bK} = \frac{abK+bt}{1+bK} = \frac{b(aK+t)}{1+bK} \end{aligned}$$

Then:

$$\frac{a-2x^*}{b} = \frac{(aK+t)}{1+bK}$$

Also:

$$2Kx^* = 2K \frac{a-bt}{2(1+bK)} = \frac{K(a-bt)}{1+bK}$$

So the LHS of the FOC becomes:

$$\begin{aligned} \frac{a-2x^*}{b} - 2Kx^* - t &= \frac{aK+t}{1+bK} - \frac{K(a-bt)}{1+bK} - t = \frac{aK+t-K(a-bt)}{1+bK} - t = \frac{aK+t-Ka+Kbt}{1+bK} - t \\ &= \frac{t+Kbt}{1+bK} - t = \frac{t(1+bK)}{1+bK} - t = 0 \end{aligned}$$

So substitution yields the identity $0=0$, confirming the algebra is correct.

e. Confirm that, for this specification of the model, profit maximization alone implies $x^{}(t) < 0$**

Starting from the first order condition (FOC) for an interior optimum,

$$F(x, t) \equiv \frac{a-2x}{b} - 2Kx - t = 0$$

Treating this as an implicit definition of $x^*(t)$, differentiate the FOC totally with respect to t :

$$\frac{\partial F}{\partial x} \frac{dx^*}{dt} + \frac{\partial F}{\partial t} = 0$$

Computing the partial derivative:

$$\frac{\partial F}{\partial x} = -\frac{2}{b} - 2k = -2\left(\frac{1}{b} + k\right), \quad \frac{\partial F}{\partial t} = 1$$

Hence:

$$\frac{dx^*}{dt} = -\frac{F_t}{F_x} = -\frac{-1}{-2\left(\frac{1}{b} + k\right)} = -\frac{1}{2\left(\frac{1}{b} + k\right)} = \frac{-b}{2(1+bk)}$$

• The sign of $\frac{dx^*}{dt}$ follows directly from the second order condition for a maximum,

$$\frac{\partial F}{\partial x} < 0 \iff \frac{1}{b} + k > 0$$

When $\frac{1}{b} + k > 0$, the denominator $2\left(\frac{1}{b} + k\right)$ is positive, while the numerator in the penultimate expression is -1, so $\frac{dx^*}{dt} < 0$. Equivalently, using the last expression, if $b > 0$ and $1 + bk > 0$ then $\frac{-b}{2(1+bk)} < 0$.

f. What is the effect on output and output price of a parallel shift in the demand curve?

We are going to consider the following parallel upward shift of the demand curve: $x = (a + \Delta) - bp$, $\Delta > 0$. This increases demand at every price but leaves the slope unchanged. Using the inverse demand function $p = \frac{a + \Delta - x}{b}$, the monopolist's profit maximizing condition under tax t is:

$$\frac{a + \Delta - 2x}{b} - 2kx - t = 0$$

Solving for the optimal output, we obtain:

$$x'(\Delta) = \frac{a + \Delta - bt}{2(1+bk)}$$

If we now differentiate with respect to Δ , we will obtain the following:

$$\frac{\delta x^*}{\delta \Delta} = \frac{1}{2(1+bk)} > 0$$

An upward shift of demand increases optimal output. The shift raises marginal revenue at every quantity, so the MR-MC intersection occurs at a higher level output.

Effect on price

The new optimal price is: $p^*(\Delta) = \frac{a + \Delta - x^*(\Delta)}{b}$

Substitute $x^*(\Delta)$:

$$p^*(\Delta) = \frac{a + \Delta}{b} - \frac{a + \Delta - bt}{2b(1 + bk)}$$

Differentiate with respect to Δ :

$$\frac{\delta p^*}{\delta \Delta} = \frac{1}{b} - \frac{1}{2b(1 + bk)} = \frac{2(1 + bk) - 1}{2b(1 + bk)} > 0$$

Because $b > 0$ and $1 + bk > 0$ by concavity. Therefore, the price also rises after a parallel upward shift of demand, though not by as much as demand shifts.

In conclusion, a parallel upward shift of the demand curve (an increase in a) increases both the monopolist's optimal output and its optimal price. Output increases because higher demand raises marginal revenue at every quantity, shifting the MR curve upward. Price rises as well, but less than proportionally, since the monopolist expands output at the new optimum.

QUESTION V3.2

Consider the technology described by $y=0$ for $x \leq 1$ and $y = \ln x$ for $x > 1$. Calculate the profit function for this technology.

Consider the production technology

$$y(x) = \begin{cases} 0 & \text{if } x \leq 1 \\ \ln(x) & \text{if } x > 1 \end{cases}$$

where $x \geq 0$ is a single input purchased at price $w > 0$, and output is sold at price $p > 0$. The firm's profit maximization problem is:

$$\Pi(p, w) = \max_{x \geq 0} \{ p y(x) - wx \}$$

This technology presents two meaningful features:

1. A production threshold, since no output is produced below one unit of input.
2. Strongly diminishing returns, because output grows only logarithmically in x .

Both elements are crucial in determining the shape of the profit function.

Below threshold

For all $x \leq 1$, output is zero, so profit is

$$\Pi(x) = -wx$$

This is strictly negative for any $x > 0$, so the optimal choice in this region is $x = 0$, which yields profit 0. Therefore the firm can always guarantee $\Pi(p, w) \geq 0$.

Above threshold

For $x > 1$, output is $y(x) = \ln(x)$. So the profit becomes:

$$\Pi(x) = p \ln(x) - wx.$$

Profit maximization in this region yields on the interior candidate:

$$\frac{\partial \Pi}{\partial x} = \frac{p}{x} - w = 0 \implies x^* = \frac{p}{w}$$

The point is feasible only when $x^* > 1$ only when $\frac{p}{w} > 1$, which requires $p > w$. Even when feasible, the firm will adopt x^* only if it yields non-negative profit, otherwise it prefers $x = 0$. Evaluating profit at the interior point:

$$\pi(x^*) = p \ln\left(\frac{p}{w}\right) - p$$

→ The firm produces only if this value is ≥ 0 :

$$p \ln\left(\frac{p}{w}\right) - p \geq 0 \iff \ln\left(\frac{p}{w}\right) \geq 1 \iff \frac{p}{w} \geq e$$

And just like that the firm engages in positive production only when both conditions hold: $p > w$ so that it's feasible in the productive region and the output price is sufficiently high relative to input cost: $p \geq ew$.

Profit function

Combining all cases, the profit function is:

$$\pi(p, w) = \begin{cases} 0 & \text{if } p \leq ew \\ p \ln\left(\frac{p}{w}\right) - p & \text{if } p > ew \end{cases}$$

This function is non-negative, convex in prices and homogeneous of degree one, as guaranteed by the standard properties of profit functions as mentioned in the profit function properties lemma or mentioned in the duality theorems. The convexity arises because $\Pi(p, w)$ is defined as the pointwise supremum of linear functions in (p, w) .

Hotelling's Lemma

Once $\Pi(p, w)$ is known, the firm's net supply/demand functions follow from Hotelling's Lemma, which states that:

- The derivative of the profit function with respect to an output price yields the supply of that output.
- The negative derivative with respect to an input price yields the conditional demand for that input.

Applying it here yields the optimal input demand:

$$x^*(p, w) = \begin{cases} 0 & \text{if } p \leq ew \\ \frac{p}{w} & \text{if } p > ew \end{cases}$$

Which coincides with the maximizer of the original problem. This consistency illustrates the Shephard-Hotelling duality principle illustrated in the given book by the professor: maximization in primal space and differentiation in dual space give the same behavioral response.

Because the technology produces zero output for all $x \leq 1$, the firm faces a dead zone where small input levels are useless. Since output grows only as $\ln(x)$ for $x > 1$ (increasing slowly) marginal productivity falls quickly. As a result, the firm will only find it worthwhile to cross the threshold and produce if the output price is sufficiently large compared to the input price.

Solving the maximization problem shows that the critical condition is $\frac{p}{w} = e$. Below this ratio, even the best feasible choice yields negative profit, so the firm optimally shuts down. Above it, the firm earns positive profit. This illustrates how the shape of the technology directly determines when production is profitable, while still fitting the standard duality properties of profit functions discussed in the notes.

QUESTION V8.14

Context

Under current tax law some individuals can save up to \$2000 a year in an Individual Retirement Account (I.R.A.) a savings vehicle that has an especially favorable tax treatment. Consider an individual at a specific point in time who has income Y , which he or she wants to spend on consumption, C , I.R.A. savings, S_1 , or ordinary savings S_2 . Suppose that the “reduced form” utility functions is taken to be:

$$U(C, S_1, S_2) = S_1^\alpha S_2^\beta C^\gamma$$

This is a reduced form since the parameters are not truly exogenous taste parameters, but also include the tax treatment of the assets, etc) The budget constraint of the consumer is given by:

$$C + S_1 + S_2 = Y$$

And the limit that he or she can contribute to the I.R.A. is denoted by L .

a. Describe the demand functions for S_1 and S_2 for a consumer for whom the limit L is not binding.

In this case, the consumer's optimal choice of S_1 is less than the maximum I.R.A contribution ($S_1^* < L$). The optimization is performed as if the I.R.A limit doesn't exist, so we only consider the budget constraint.

We first use the budget constraint to eliminate one variable, typically C , from the utility function. This simplifies the three variable problem into a two variable one. From the budget constraint: $C = Y - S_1 - S_2$ Substituting into the utility function, we maximize the respect to S_1 and S_2 :

$$\max_{S_1, S_2} U = S_1^\alpha S_2^\beta (Y - S_1 - S_2)^\gamma$$

Then we use the standard method of setting the partial derivatives of the objective function with respect to S_1 and S_2 to zero:

- Partial derivative with respect to S_1 :

$$\frac{\partial U}{\partial S_1} = \alpha S_1^{\alpha-1} S_2^\beta C^\gamma - \gamma S_1^\alpha S_2^\beta C^{\gamma-1} = 0$$

→ Dividing by $S_1^\alpha S_2^\beta C^{\gamma-1}$ gives the simplification:

$$\frac{\alpha}{S_1} - \frac{\gamma}{C} = 0 \Rightarrow \frac{S_1}{C} = \frac{\alpha}{\gamma}$$

• Partial derivative with respect to S_2 :

$$\frac{\delta U}{\delta S_2} = \beta S_1^\alpha S_2^{\beta-1} C^\gamma - \gamma S_1^\alpha S_2^\beta C^{\gamma-1} = 0$$

→ Dividing by $S_1^\alpha S_2^{\beta-1} C^{\gamma-1}$ gives the simplification:

$$\frac{\beta}{S_2} - \frac{\gamma}{C} = 0 \Rightarrow \frac{S_2}{C} = \frac{\beta}{\gamma}$$

From the simplified conditions, we see that the optimal savings and consumption levels are proportional to the utility exponents α, β, γ :

$$S_1 = \frac{\alpha}{\gamma} C \text{ and } S_2 = \frac{\beta}{\gamma} C$$

Now we substitute these back into the main budget constraint, $C + S_1 + S_2 = Y$:

$$C + \frac{\alpha}{\gamma} C + \frac{\beta}{\gamma} C = Y \rightarrow C \left(1 + \frac{\alpha + \beta}{\gamma} \right) = Y$$

$$C \left(\frac{\gamma + \alpha + \beta}{\gamma} \right) = Y$$

Let $K = \alpha + \beta + \gamma$ be the sum of the exponents. Solving for C we are left with $C' = \frac{\gamma}{K} Y$

The demand functions for S_1 and S_2 are then derived by substituting C' back into the proportionality equations:

- Demand for $S_1 \rightarrow$ I.R.A. savings:

$$S_1^*(Y) = \frac{\alpha}{\gamma} C' = \frac{\alpha}{\gamma} \left(\frac{\gamma}{K} Y \right) = \frac{\alpha}{K} Y$$

- Demand for $S_2 \rightarrow$ ordinary savings:

$$S_2^*(Y) = \frac{\beta}{\gamma} C' = \frac{\beta}{\gamma} \left(\frac{\gamma}{K} Y \right) = \frac{\beta}{K} Y$$

In conclusion, when the I.R.A. limit is not binding the consumer's demand for S_1 and S_2 are a constant fraction of their total income Y , determined by the ratio of their respective utility exponent to the total sum of all exponents, K .

b. Describe the demand functions for S_1 and S_2 for a consumer for whom the limit L is binding.

In this scenario, the consumer's unconstrained optimal choice for S_1 from case 1 ($\frac{\alpha}{K}Y$) is greater than the I.R.A. limit (L). Since the customer wishes to save more in the I.R.A. but cannot, they must choose the maximum legal amount for $S_1 \rightarrow S_1^* = L$. This new restriction converts the problem into a simpler optimization with a fixed amount for S_1 . The budget constraint becomes the following:

$$C + L + S_2 = Y \Rightarrow C = Y - L - S_2$$

The consumer is now maximizing utility by choosing only S_2 (and implicitly C), with S_1 fixed at L :

$$\max_{S_2} U = L^\alpha S_2^\beta (Y - L - S_2)^\gamma$$

Let $Y' = Y - L$ be the remaining discretionary income to be allocated between C and S_2 . The consumer is maximizing $U \propto S_2^\beta C^\gamma$ subject to $C + S_2 = Y'$ where L^α term is a constant that doesn't affect the optimal ratio.

We maximize the function by taking the partial derivative with respect to S_2 and setting it to zero:

$$\frac{\partial U}{\partial S_2} = L^\alpha \left(\beta S_2^{\beta-1} C^\gamma - \gamma S_2^\beta C^{\gamma-1} \right) = 0$$

→ Dividing by $L^\alpha S_2^{\beta-1} C^{\gamma-1}$ gives the simplified condition:

$$\frac{\beta}{S_2} - \frac{\gamma}{C} = 0 \Rightarrow \frac{S_2}{C} = \frac{\beta}{\gamma}$$

Similar to the unconstrained case, we find the proportional relationship: $S_2 = \frac{\beta}{\gamma}C$. We substitute this back into the new, reduced budget constraint, $C + S_2 = Y'$:

$$C + \frac{\beta}{\gamma}C = Y' \Rightarrow C\left(\frac{\gamma + \beta}{\gamma}\right) = Y'$$

Let $K' = \beta + \gamma$ be the sum of the exponents on the remaining choices. Solving for C :

$$C'' = \frac{\gamma}{K'} Y' = \frac{\gamma}{\beta + \gamma} (Y - L)$$

The demand function for S_2 is derived by substituting C'' back, leaving us the following:

- Demand for $S_1 \rightarrow$ I.R.A. savings:

$$S_1''(Y, L) = L$$

- Demand for $S_2 \rightarrow$ ordinary savings:

$$S_2''(Y, L) = \frac{\beta}{\gamma} C'' = \frac{\beta}{\gamma} \left(\frac{\gamma}{\beta + \gamma} (Y - L) \right) = \frac{\beta}{\beta + \gamma} (Y - L)$$

In summary, when the I.R.A. limit is binding S_1 is fixed at L . The consumer treats $Y - L$ as the remaining income, and allocates S_2 as a constant fraction of this remaining income, where the fraction is determined by the ratio of β to the sum of the remaining exponents $\beta + \gamma$.